

Welcome to Colab!

```
# Import Libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
sns.set_style("whitegrid")
```

```
# 1. LOAD DATASET
```

```
from google.colab import files
uploaded = files.upload()
```

Choose files train.csv
train.csv(text/csv) - 61194 bytes, last modified: 11/12/2019 - 100% done
Saving train.csv to train (1).csv

```
df = pd.read_csv("/content/train.csv")

print("First 5 Rows:")
print(df.head())

print("\nDataset Info:")
print(df.info())
```

```
First 5 Rows:
  PassengerId  Survived  Pclass \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                Name    Sex  Age  SibSp \
0                        Braund, Mr. Owen Harris   male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0      1
2                        Heikkinen, Miss. Laina   female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)   female  35.0      1
4                        Allen, Mr. William Henry   male  35.0      0

   Parch    Ticket   Fare Cabin Embarked
0      0  A/5 21171   7.2500   NaN        S
1      0    PC 17599  71.2833   C85        C
2      0 STON/O2. 3101282   7.9250   NaN        S
3      0   113803   53.1000  C123        S
4      0   373450   8.0500   NaN        S
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

2. DATA CLEANING

```
print("\nMissing Values Before Cleaning:")
print(df.isnull().sum())
```

Missing Values Before Cleaning:

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

Fill missing Age values with mean

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

Fill missing Embarked Values with mode

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

Drop Cabin Column (too many missing values)

```
df.drop('Cabin', axis=1, inplace=True, errors='ignore')
```

Missing values After cleaning

```
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())
```

Missing Values After Cleaning:

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64
```

3. FEATURE ENGINEERING

Create Family Size

```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

Create Age Groups

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[0, 12, 19, 35, 60, 100],
    labels=['Child', 'Teen', 'Young Adult', 'Adult', 'Senior']
)
```

4. ANALYSIS

```

print("\n=====")
print("SURVIVAL RATE BY AGE GROUP")
print("=====")

age_survival = df.groupby('AgeGroup')['Survived'].mean() * 100
print(age_survival)

print("\n=====")
print("SURVIVAL RATE BY EMBARKATION PORT")
print("=====")

embark_survival = df.groupby('Embarked')['Survived'].mean() * 100
print(embark_survival)

print("\n=====")
print("SURVIVAL RATE BY FAMILY SIZE")
print("=====")

family_survival = df.groupby('FamilySize')['Survived'].mean() * 100
print(family_survival)

```

```

=====
SURVIVAL RATE BY AGE GROUP
=====

```

```

AgeGroup
Child      57.971014
Teen       41.052632
Young Adult 35.294118
Adult      40.000000
Senior     22.727273
Name: Survived, dtype: float64

```

```

=====
SURVIVAL RATE BY EMBARKATION PORT
=====

```

```

Embarked
C      55.357143
Q      38.961039
S      33.900929
Name: Survived, dtype: float64

```

```

=====
SURVIVAL RATE BY FAMILY SIZE
=====

```

```

FamilySize
0      30.353818
1      55.279503
2      57.843137
3      72.413793
4      20.000000
5      13.636364
6      33.333333
7       0.000000
10     0.000000
Name: Survived, dtype: float64

```

```

/tmp/ipykernel_6608/2302420006.py:5: FutureWarning: The default of observed=False is deprecated and will be changed to True in a
age_survival = df.groupby('AgeGroup')['Survived'].mean() * 100

```

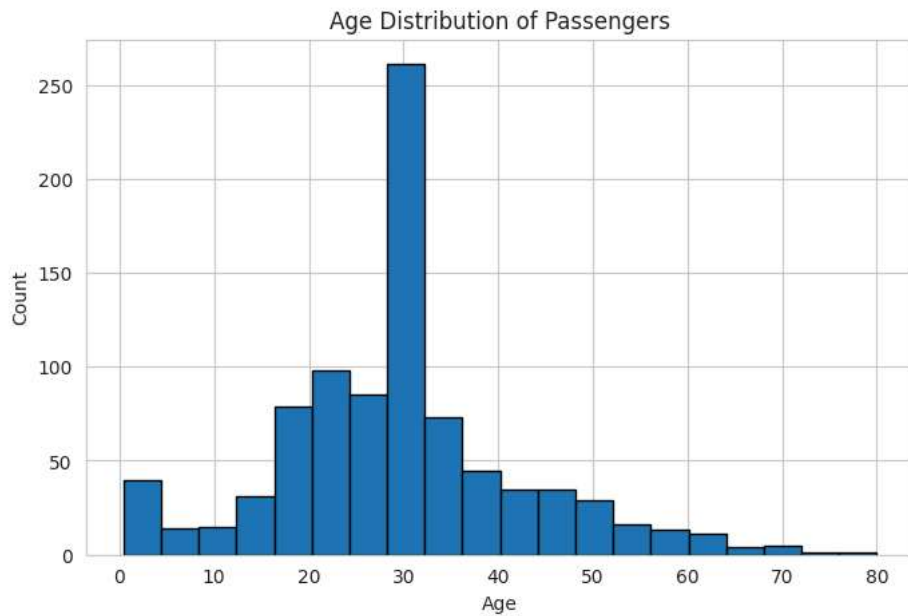
5. VISUALIZATION 1

Age Distribution Histogram

```

plt.figure(figsize=(8,5))
plt.hist(df['Age'], bins=20, edgecolor='black')
plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()

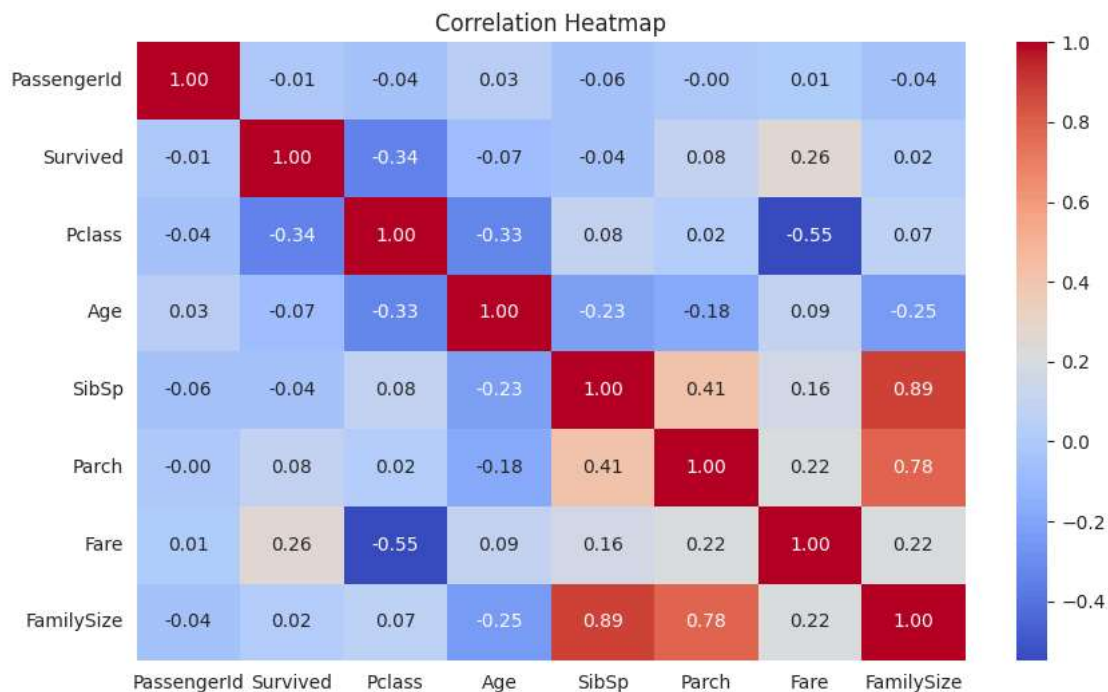
```



```
# 6. VISUALIZATION 2
# HEATMAP OF CORRELATIONS
```

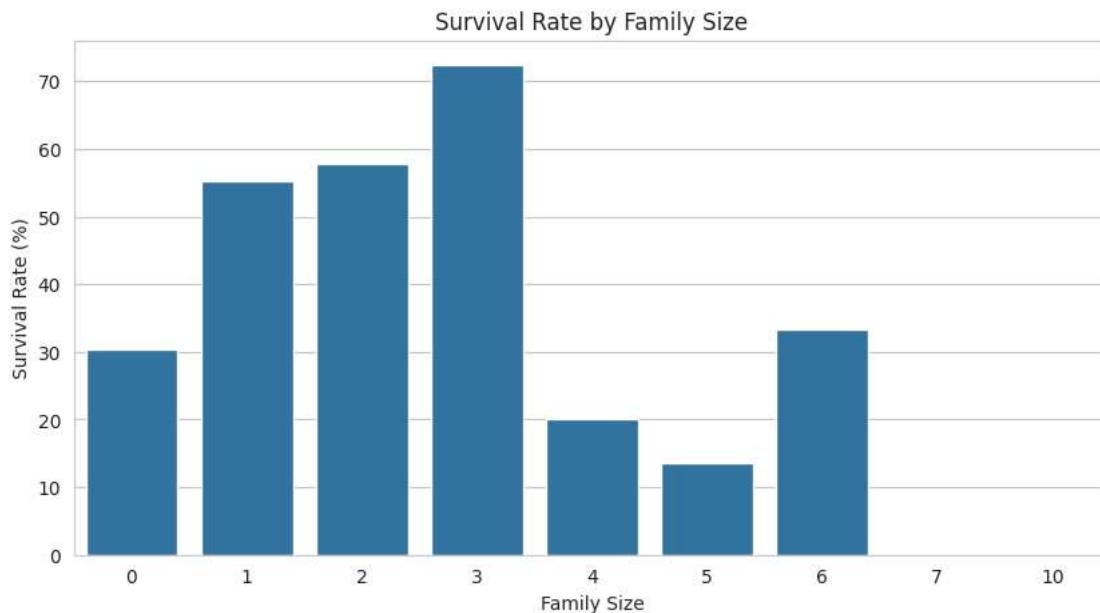
```
numeric_df = df.select_dtypes(include=['number'])

plt.figure(figsize=(10,6))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)
plt.title('Correlation Heatmap')
plt.show()
```



```
# 7. VISUALIZATION 3
# SURVIVAL BY FAMILY SIZE
```

```
plt.figure(figsize=(10,5))
sns.barplot(
    x=family_survival.index,
    y=family_survival.values
)
plt.title('Survival Rate by Family Size')
plt.xlabel('Family Size')
plt.ylabel('Survival Rate (%)')
plt.show()
```



```
# 8. ADDITIONAL VISUALIZATION
# SURVIVAL RATE BY AGE GROUP
```

```
plt.figure(figsize=(8,5))
sns.barplot(
    x=age_survival.index,
    y=age_survival.values
)
plt.title('Survival Rate by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Survival Rate (%)')
plt.show()
```

